Decadal Estimates of Surface Mass Balance for Glaciers in Chandra Basin, Western Himalayas, India—A Geodetic Approach

RAAJ Ramsankaran, Ankur Pandit and Avinash Parla

Abstract The Himalayas, which has the highest concentration of glaciers outside the poles, form a huge reservoir of fresh water. These glaciers are not only important because of the contribution they make to our water resources, but also because they are a great help in understanding the effects of climate change. However, monitoring even a few of these glaciers through field techniques is a tedious process due to the tough and unapproachable terrain of the Himalayas. In such situation, remote sensing techniques become a boon. In this article, one such widely used remote sensing-based technique called digital elevation model (DEM). Differencing or geodetic approach available for estimating glacier surface mass balance (SMB) is discussed. However, the accuracy of the estimated SMB using geodetic approach is highly dependent on the accuracy of the DEMs used, which ultimately decides whether this approach can be used for seasonal, annual or decadal studies. To get a better understanding of this approach, this article gives a brief overview of errors present in DEMs and how to correct it before estimating SMB by demonstrating it over 65 glaciers located in the Chandra basin, Lahaul–Spiti district of Himachal Pradesh, India. The obtained results indicate that the Chandra basin has lost 5.01 ± 2.48 Gt of ice mass during the course of 14 years (1999–2013), which gives a clear signal on the effect of climate change occurred over the study area and how climate change could cause imbalance in the health of glacier(s). Considering the advantages of this approach and recent developments in high-resolution DEM data acquisition as well as the need for such accurate mass balance estimates across Himalayas, it is therefore suggested that this

R. Ramsankaran (✉)
Department of Civil Engineering, Indian Institute of Technology Bombay, Room No. 202, Mumbai 400076, India
e-mail: ramsankaran@civil.iitb.ac.in

R. Ramsankaran · A. Pandit
Interdisciplinary Programme in Climate Studies, Indian Institute of Technology Bombay, Mumbai 400076, India
e-mail: ankkurpandit@gmail.com

A. Parla
Water Resources Division, Department of Civil Engineering, Indian Institute of Technology Bombay, Mumbai 400076, India
e-mail: p.avinash283@gmail.com

© Springer Nature Singapore Pte Ltd. 2019

C. Venkataraman et al. (eds.), *Climate Change Signals and Response*,
https://doi.org/10.1007/978-981-13-0280-0_7

p.avinash283@gmail.com

modelling approach should be extended to other basins to get the current status of the health of the glaciers across Himalayas.

1 Introduction

The Hindu Kush Himalayan (HKH) region has the highest concentration of glaciers outside the poles, which makes them a huge source of fresh water for the developing world such as Afghanistan, Pakistan, India, Tibet, Nepal, Bhutan to which they feed. One-third of the humanity on this planet is directly or indirectly dependent on nine major river systems to which headwater is provided by the Himalayan glaciers distributed along the Himalayan arc. Water from these rivers has a significant socio-economic effect on more than 700 million people (Immerzeel et al. 2010). The runoff coming from these glaciers not only affect the drinking water supplies, but also the hydropower projects, biodiversity, industries, agriculture and day-to-day life of dependent individuals among others. HKH glaciers are not only important because of the contribution they make to our water resources, but also because they are widely recognized to be sensitive climatic indicators (Houghton et al. 2001). As compared to the polar region, the HKH glaciers receive more heat because of its location which is closer to the Tropic of Cancer, and hence, they are very sensitive towards climate variability or rising temperature both at regional and global levels.

Out of the entire HKH region, Indian Himalayan region is significantly important because of its rich biodiversity and its influence in Indian summer monsoon, etc. As per one of the latest report by Geological Survey of India (GSI) about 9575 glaciers with an estimated area of 37,466 km^2 exist in Indian Himalayas (Sangewar and Shukla 2009). The three major river systems in India originating from these glaciers, i.e. the Indus, the Ganges and the Brahmaputra, provides approximately 50% of the annual usable water resources (690 km^3) of the country (DST 2012). Out of this, a significant contribution is coming from glacier and seasonal snowmelt especially in Indus basin. Considering the larger extent of water utilization, understanding the changes in these glaciers is vital, especially, for the current and future government policies for the regional water resources management.

The Inter-Governmental Panel on Climate Change (IPCC) in its fourth assessment report (IPCC 2007) highlighted the melting of ice mass and glaciers worldwide as one of the most prominent indicators of climate change. IPCC fifth assessment report on climate impacts (IPCC 2014) says that Himalayan glaciers would shrink by 45% by 2100, if Earth's average surface temperature rose by 1.8 °C. Under a far warmer scenario of 3.7 °C, the reduction would be 68%. IPCC (2007) highlighted that the global mean surface air temperature is increased by 0.74 °C over the last century in the Himalayas. In agreement with the observations of IPCC (2007) and projections of IPCC (2014) reports, the temperature trends profile of the north-west Himalayan region has already shown a significant warming of 1.6 °C (Bhutiyani et al. 2007), whereas 0.98 °C of temperature warming is observed in the western Himalayan regions (Dash et al. 2007). An accelerated warming after

p.avinash283@gmail.com

1972 was observed in the north-west and western Himalayan regions (Dash et al. 2007). Another study conducted using microwave satellite measurements from 1979 to 2007 also highlighted that annual mean warming over the whole HKH region is accelerated by 0.21 ± 0.08 °C/decade with maximum warming localized over the western Himalaya, i.e. 0.26 ± 0.09 °C/decade (Gautam et al. 2010). In the Karakoram region, a very interesting contrast is observed by Dimri and Dash (2010) where they found a decrease in winter temperatures in the accumulation zone whereas an increase in winter temperatures in the ablation zone of the Siachen Glacier between year 1984 and 2006. The contrast in temperature is more likely to be directed by local dynamics rather than by regional or global trends (Singh et al. 2011). It is expected that changes in temperature due to climate change might affect the pattern of precipitation. Dimri and Dash (2012) reported decreasing trends in winter precipitation (December–February) in the western Himalaya for 1975–2006. The temperature warming in the western Himalayas has caused an increase in the ablation area of the glaciers by shifting snowline towards the higher elevations. This upward shifting of snowline has gathered pace in the past few years.

Under the climate changing scenario, the most crucial to the survival of a glacier is its mass balance, i.e. the difference between snow/ice accumulation and ablation. A glacier with a sustained negative balance is out of equilibrium and will retreat, whereas a glacier with a sustained positive balance is out of equilibrium and will advance. As climate change may cause variations in both temperature and snowfall, changes in mass balance can control a glacier's long-term behaviour. There is an overwhelming evidence that the Himalayan glaciers especially in western region are not in the best of health and most of them are losing mass (e.g. Vijay and Braun 2016). In another study conducted by Kulkarni et al. (2007) on 466 glaciers of the western Himalayas highlighted that there has been a 21% reduction in the glacierized area, i.e. from 2077 km^2 in 1962 to 1628 km^2 in 2004. Smaller glaciers of less than 1 km^2 have reduced in area by as much as 38% compared to 12% retreat of the larger glaciers. While most of the glaciers are losing mass at a rate similar to glaciers at other places, Karakoram Himalayas emerge out as an exception with some studies suggesting mass gain (Bolch et al. 2012). The region of western Himalayas, in particular, is extremely interesting to investigate because the region is influenced by two major climatic systems, i.e. the mid-latitude Westerlies and the south Asian summer monsoon.

One of the methods to estimate mass balance of a glacier in field is through glaciological method, which requires extensive field surveys over the rugged and tough terrain of the Himalayan glaciers. This traditional stake-based measurement is still carried to collect long-term information regarding the glacier health (Dobhal et al. 2004; Kumar et al. 2009; Azam et al. 2012). In this approach, the rate of data acquisition is slow and expenses for logistics and labour can be high in maintaining observational networks at the higher elevations. Also, extreme weather and terrain conditions restrict the long-term field mass balance measurement in the Himalayan region. In India, there are approximately 25 research institutes/organizations are currently involved in glacier-related research. However, considering the extent of the Himalayan region, field mass balance observations are available only for about 12 glaciers (DST 2012) out of which most of them are not continuous. Till date, most of

p.avinash283@gmail.com

the glaciers in the Indian Himalayan region are untouched due to their inaccessibility and/or the shortage of manpower with proper training. This largely explains the very limited availability of mass balance data in the Himalaya.

To overcome the dependency of field-based mass balance measurements, a very concerted research effort is required to develop alternate ways of securing this important data. Under such circumstances, remote sensing technology seems to be a very promising tool for temporal monitoring of glacier health. Images acquired from different platforms such as satellite, aircrafts using sensors that operate in different spectral regions such as visible, infrared, microwave have been widely used to study glacier mass balance. (e.g. Gardelle et al. 2012; Kääb et al. 2012; Vincent et al. 2013; Mernild et al. 2013; Rabatel et al. 2016).

Among the available methods for estimating mass balance from remote sensing data, the geodetic method, also known as digital elevation model (DEM) differencing approach has been widely used for large-scale studies by many researchers (e.g. Berthier et al. 2004; Gardelle et al. 2012; Kääb et al. 2012; Vincent et al. 2013). Based on historical topographic maps and/or high-resolution DEMs derived from Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) (e.g. Rivera and Cassassa 1999; Kääb 2007), Satellite Pour l'Observation de la Terre (SPOT) imagery (Berthier et al. 2007), Shuttle Radar Topography Mission (SRTM) (e.g. Surazakov et al. 2006; Racoviteanu et al. 2007), high-resolution Advanced Land Observing Satellite (ALOS) Panchromatic Remote sensing Instrument for Stereo Mapping (PRISM) and Corona (Narama et al. 2009), laser altimetry (Arendt et al. 2002), ERS, Radarsat (Dedieu et al. 2003) and TerraSAR-X Add-On for Digital Elevation Measurement (TanDEM-X) (e.g. Vijay and Braun 2016) glacier surface mass balances can be estimated via estimating glacier thickness change. This method measures the elevation changes over time ($\delta h/\delta t$) from various available DEMs constructed over a glacier surface. Historical topographic information is often used to construct elevations models which are subtracted from more recent DEMs constructed from remote sensing imagery either on a pixel-by-pixel basis or as average elevation change to obtain difference maps. The availability of spaceborne elevation data of glaciers present in the remote location increases the capability to quantify glacier changes. However, the accuracy of the geodetic method has several issues. It highly depends on the accuracy of the DEMs used, discrepancies in the scale of resolution of DEMs and the assumption of the density of lost/gained material (snow/firn/ice). This approach is suitable for the estimation of long-term variation (5–10 years) rather than yearly variation in the glacier elevation because of the low vertical accuracy of generated DEMs. It is important to highlight that many earlier studies use the DEMs without correcting for biases between them as pointed out by, e.g. Rignot et al. (2003), Sund et al. (2009), Muskett et al. (2009). This may lead to wrong estimates of glacier volume changes (Berthier et al. 2010). The geodetic surface mass balance without the correction of potential biases has been compared with the surface mass balance estimated from direct glaciological field measurements in a number of studies, and the differences are often found to be large. (e.g. Krimmel 1999; Østrem and Haakensen 1999; Rolstad et al. 2009). Recently, there are

p.avinash283@gmail.com

many approaches proposed for correcting the bias in DEMs and tested successfully in various locations (e.g. Schiefer et al. 2007; Van Niel et al. 2008).

Considering the above lacuna in the mass balance information across Himalayas and the recent developments in geodetic approach, the motivation behind the current study is to demonstrate the applicability of bias-corrected DEM Differencing approach to provide an accurate/reliable surface mass balance. For experimental purpose, the DEM differencing approach with appropriate corrections for DEM bias has been demonstrated over 65 glaciers located in the Chandra basin of western Himalayan region in India to get decadal glacier surface mass balance estimates.

Following this introduction (Sect. 1), this chapter discusses details of the study site chosen and the datasets used in the current study in Sect. 2. In Sect. 3, the methodology adopted to investigate the decadal surface mass balance is discussed. Later on, in Sect. 4, the inferences drawn from the current study are discussed followed by the conclusion and recommendations in Sect. 5.

2 Study Area and Datasets Used

The study was conducted over 65 glaciers of Chandra basin located in the western Himalaya, Lahaul–Spiti valley of Himachal Pradesh, India. These 65 glaciers are selected based on their size (<0.5 km^2) because small glaciers have high uncertainty in glacier outlines. The location map of Chandra Basin is shown in Fig. 1.

The basin is having an area of ~2381 km^2 (Pandey et al. 2017). This basin is a sub-basin of Chenab basin and ranging from an elevation range of 2400 m a.s.l. and 6400 m a.s.l. Glaciers from this basin feed the Chandra River—a major tributary of the Chenab river system. Geomorphologically, the area represents highly rugged, inaccessible terrain with high mountains and deeply dissected valleys (Sharma et al. 2016). Bara Shigri Glacier with an area of 136 km^2 is the largest glacier in this basin. This basin falls in the monsoon-arid transition zone and is alternately influenced by Indian Summer Monsoon during summer and mid-latitude westerlies during winter (Wagnon et al. 2007; Bookhagen and Burbank 2010). High wet precipitation is recorded in summer season (July–September), whereas winter season (November–February) experiences a significant amount of solid precipitation due to the influence of the westerlies (Sharma et al. 2013). This basin is characterized by heavy and dry snowfall, fairly cold temperatures with strong wind action (Sharma and Ganju 2000; Negi et al. 2013).

The current study has utilized SRTM C-band DEM and TanDEM-X DEM for estimating decadal surface mass balance of the study area. SRTM is a joint program between NASA and National Geo-spatial intelligence Agency (NGA) to map the globe in three dimensions. Flown on 11–22 February 2000, SRTM had the first space-borne single-pass SAR interferometer. It yielded single-pass interferometry data, C- and X-bands, of land between 60N and 54S for processing into global DEMs. The InSAR antenna configuration (precise measurement of relative antenna location with laser rangers, star trackers and in-orbit Global Positioning System—GPS) resulted

p.avinash283@gmail.com

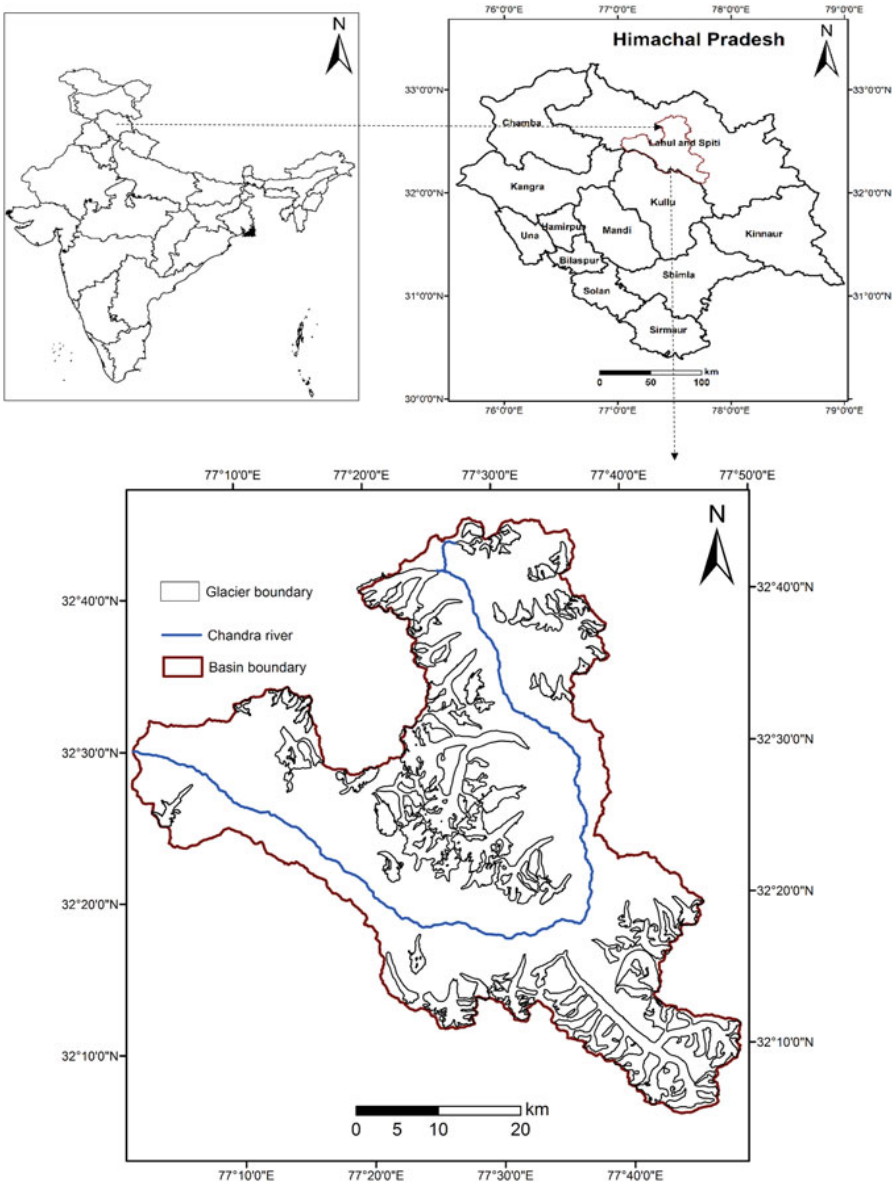


Fig. 1 Map of the study area showing the locations of the selected 65 glaciers in Chandra river basin, Lahaul–Spiti valley, Himachal Pradesh, India

in elimination of mapping errors characteristic of repeat-pass satellite InSAR techniques. The nominal vertical accuracy of SRTM is 6 m relative, 16 m absolute, and the nominal horizontal accuracy is 15 m relative, 20 m absolute, at the 90% confidence level (Muskett et al. 2003). Likewise, high-resolution TanDEM-X (TDX) satellite

p.avinash283@gmail.com

data obtained under the science proposal XTI_GLAC7043 project has been utilized for generation of another DEM. For this purpose, two raw TDX data was ordered to create a high-resolution DEM (10 × 10 m resolution) of the Chandra basin covering the selected 65 glaciers. The two TDX tiles were acquired on 27th October 2013 in HH polarization with normal baseline of 182.66, 182.72 m and incidence angle of 34.39°, 34.56°, respectively. Both the tiles were acquired on descending pass. SARscape 5.0 licensed software was used for generating DEM from TanDEM-X satellite data. For the current study, the outlines of the study glaciers were obtained from the Randolph Glacier Inventory 6.0 (https://www.glims.org/RGI/) and wherever required, minor corrections have been done using high-resolution Google Earth images.

3 Methodology

The approach adopted for estimating decadal glacier specific Surface Mass Balance (SMB) using bias-corrected DEMs is illustrated in Fig. 2. The biases corrected in this study are (i) the bias due to penetration of microwave signals especially in the C-band of SRTM, (ii) the bias caused due to mis-registration of the two DEMs, (iii)an elevation-dependent bias, which is observed when the DEMs are subtracted and plotted with the altitudes (Berthier et al. 2006, 2007).

Mis-registration bias occurs when two different DEMs of the same terrain surface are not accurately aligned, showing a characteristic bias in elevation difference. However, the reason of the elevation-dependent bias is not clearly known but few studies have suggested some reasons like varying resolution (Paul 2008), more penetration of SRTM C-band at higher elevations (Berthier et al. 2006). Either way, an elevation-dependent bias is extremely significant for estimating glacier volume changes because a glacier and its mass balance vary predominantly with elevation. In this study, we consider that the bias in elevation is caused only due to varying resolutions of the two DEMs used.

The TDX DEM was generated using SAR interferometry technique as given in Pandit et al. (2014) with an overall vertical accuracy of 7.41 m (Ramsankaran et al. 2018). Thus obtained 10-m TDX DEM was then resampled to 30 m to match the SRTM DEM. Due to the fact that C-band radar penetration of SRTM can reach up to 10 m in snow and ice, SRTM DEM thus leading to biased estimates of glacier elevation changes. Therefore, the C-band snow penetration correction as suggested by Kääb et al. (2012) was applied over the SRTM DEM. Then, both the DEMs were further utilized in co-registration algorithm suggested by Nuth and Kääb (2011). For more details about the algorithm, readers can refer to Nuth and Kääb (2011).

Later on, following Gardelle et al. (2012) the bias appeared due to varying DEMs resolution are corrected on and off glacier based on the relation established between the elevation difference and terrain maximum curvature for the stable terrain of the glacier. Maximum curvature is obtained from the high-resolution TDX DEM using its native resolution of 10 m. Once the biases in DEMs are corrected, the ice thickness

p.avinash283@gmail.com

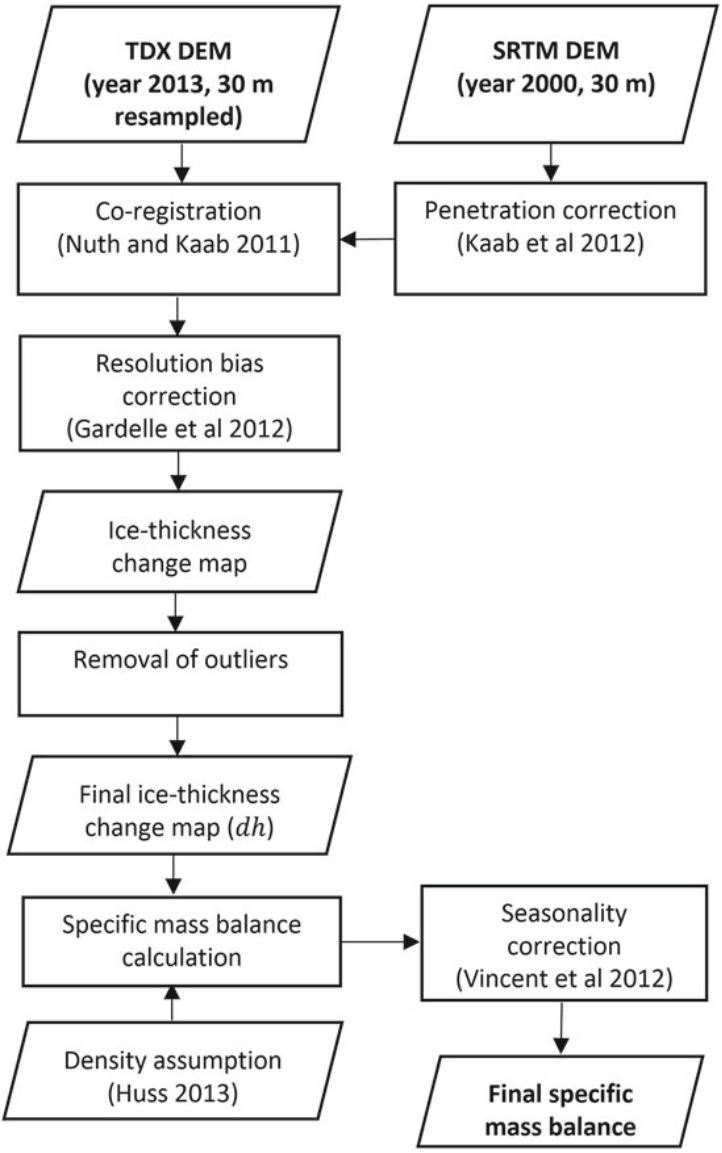


Fig. 2 Methodology flow chart for estimating glacier specific SMB using bias-corrected DEMs

change map for all the 65 glaciers was generated. Then following Berthier et al. (2004), the resultant elevation changes on a glacier that exceed ±100 m are discarded initially, which are surely considered to be outliers. Next, in order to eliminate the further outliers present in the thickness change, we have discarded pixels for which the height difference differed by more than three standard deviations (±100 m) from the mean. Then, using a 3 × 3 averaging filter the excluded pixels were filled. The

p.avinash283@gmail.com

filled ice thickness map was then converted into specific SMB using Eq. 1 in which the density of ice is assumed to be 850 ± 60 kg/m^3 as adopted by Huss (2013).

$$M = \frac{\int_A dh}{A} \times \frac{\rho_{\text{ice}}}{\rho_{\text{water}}} \tag{1}$$

where M = specific SMB (m w.e), dh = Elevation difference (m), ρ_{ice} = Density of ice (kg/m^3), ρ_{water} = Density of water (1000 kg/m^3), A = Glacier surface area (m^2).

To estimate specific SMB rate seasonal corrections are to be applied which would restore the specific SMB to a value appropriate to the beginning or end of the balance year, or to the date of the other survey. Therefore, a seasonality correction to cover 14 full years from October 1999 to October 2013 was applied between October 1999 and February 2000 as suggested by Vincent et al. (2013). The possible mass changes during this 4-month period were derived from the mean winter mass balance of Chhota Shigri Glacier given in Azam et al. (2016), which comes out to be about 0.15 m w.e per winter month. Finally, uncertainty in the estimated specific SMB rate has also been calculated using Eq. 2 obtained through error propagation method. While calculating the uncertainty in the specific SMB rate, the errors in elevation difference, density (850 ± 60 kg/m^3) and area of the glacier (5% relative uncertainty is assumed) are considered as sources of uncertainty.

$$\frac{\nabla M}{M} = \sqrt{\left(\frac{\nabla \overline{dh}}{\overline{dh}}\right)^2 + \left(\frac{\nabla \rho_{\text{ice}}}{\rho_{\text{ice}}}\right)^2 + \left(\frac{\nabla A}{A}\right)^2} \tag{2}$$

where ∇ represents the uncertainty in estimates.

4 Results and Discussion

Following the methodology specified in Fig. 2, the ice thickness changes and specific SMB rates in the selected 65 glaciers of Chandra basin are calculated. Spatial distribution of the obtained ice thickness change for each of the selected 65 glaciers is presented in Fig. 3. The colour-coded ice thickness change map (Fig. 3) clearly shows that all the glaciers are thinning in most of the parts and a little thickening is observed at accumulation zones of each glacier. The thinning and thickening are found to be varying locally up to 60 and 40 m, respectively.

Specific SMB rate of the selected 65 glaciers of Chandra basin obtained using the bias-corrected DEMs are shown in Fig. 4 along with the area of each glacier. It is found that the specific SMB rate of different glaciers in Chandra basin range from -0.83 ± 0.407 m w.e.y^{-1} to -0.14 ± 0.07 m w.e.y^{-1}. The most negative specific SMB rate of -0.83 ± 0.407 m w.e.y^{-1} is obtained for the G077662E32308N RGI glacier. The mean specific SMB rate of all the glaciers in Chandra basin is about -0.61 ± 0.29 m w.e.y^{-1}. In comparison with the whole basin, it is observed that

p.avinash283@gmail.com

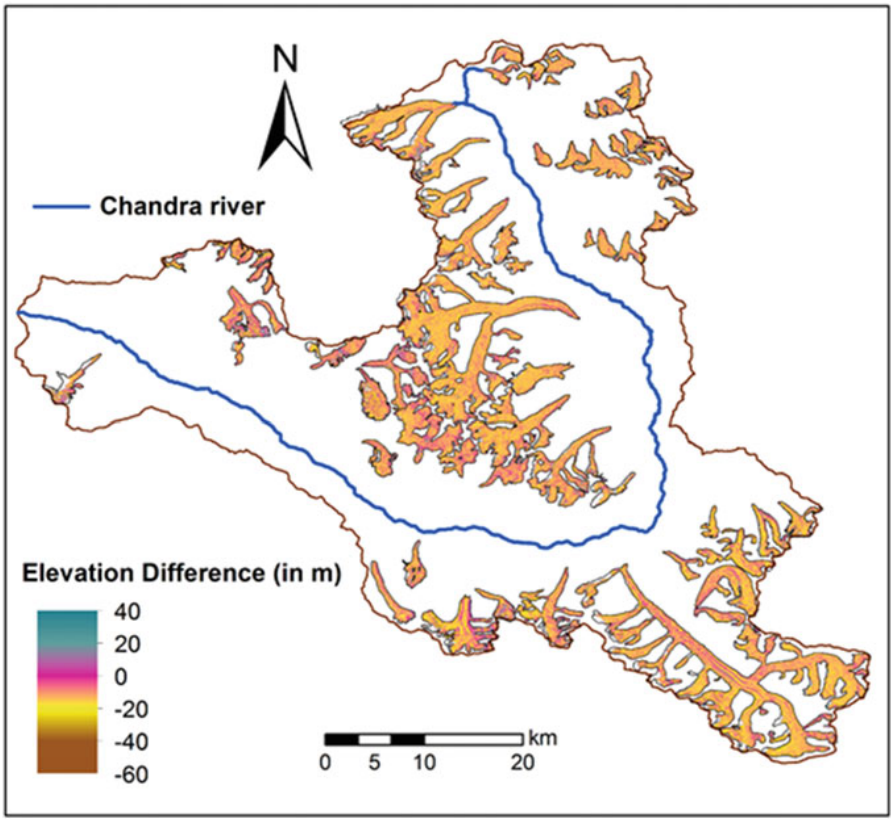


Fig. 3 Map of ice thickness changes of the selected 65 glaciers in the Chandra basin between February 2000 and October 2013

the mass loss rate is lower in Chhota Shigri Glacier (a benchmark glacier), which is estimated to be -0.55 ± 0.26 m w.e.y^{-1}. On the other hand, the surface mass loss rate in large glaciers such as Samudra Tapu and Bara Shigri is found to more than the rest of the region, which are estimated to be -0.69 ± 0.33 m w.e.y^{-1} and -0.66 ± 0.32 m w.e.y^{-1} , respectively.

From the obtained results, it is found that the basin lost around 5.01 ± 2.48 Gt of ice in the past 14 years. Out of which 37.5% was lost from some of the large glaciers such as Bara Shigri and Samudra Tapu glaciers, which indicates that the mass loss is highly dependent on the area of the glacier. To understand how the mass loss is varying in different altitude bands a hypsometric distribution of mass loss is shown in Fig. 5 where mass lost is plotted against the altitude bands along with glaciated area of each altitude band. From Fig. 5, it is inferred that about 3.31 ± 1.66 Gt ice have lost in the elevation band of 5000–5600 m, which is almost 65% of the total loss. The glacier area in lower altitude bands lost less ice mass as their area is less. However, it seen that specific mass loss (defined as mass loss per area) at lower

p.avinash283@gmail.com

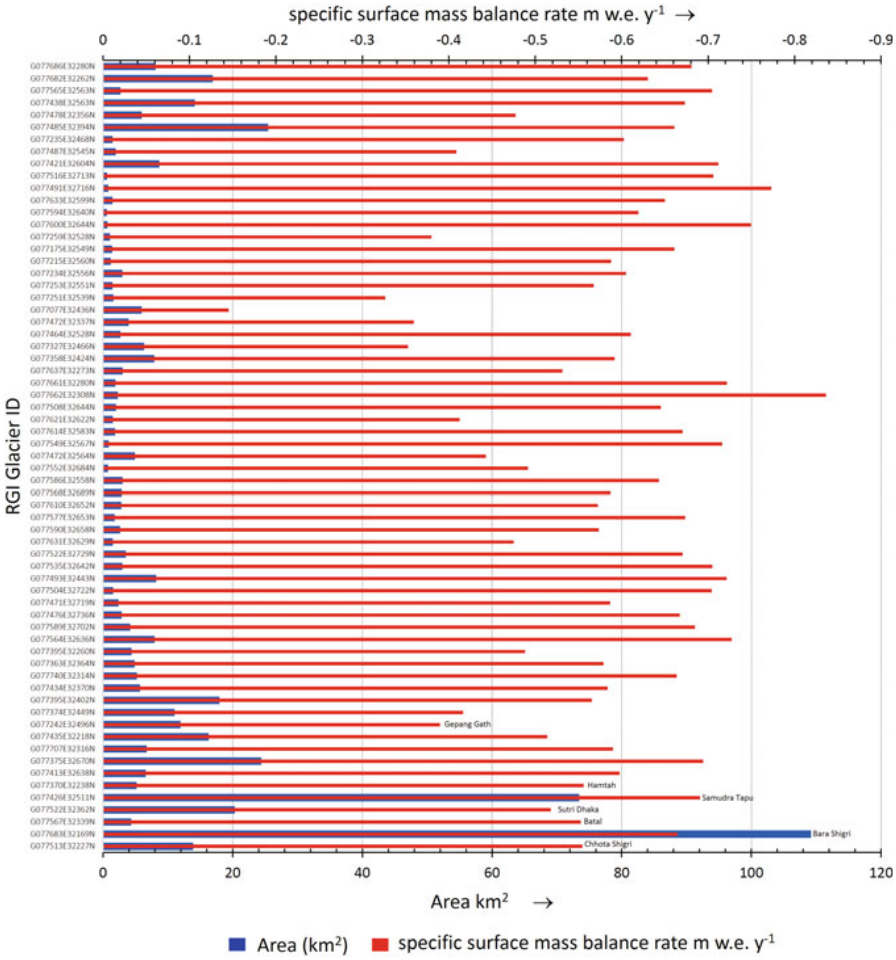
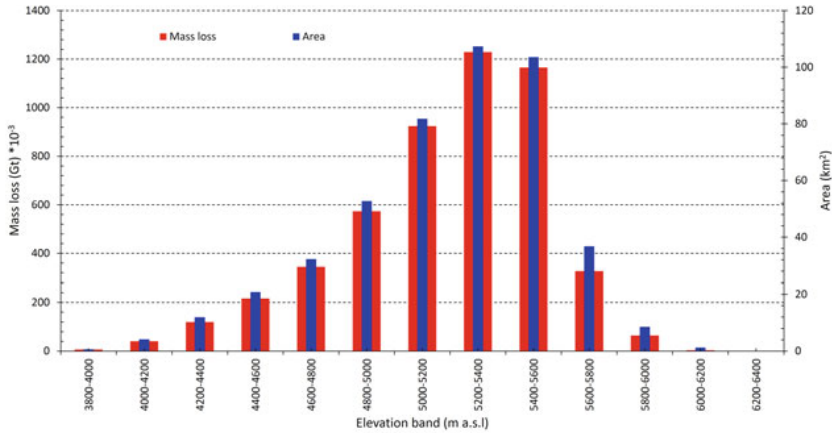


Fig. 4 Specific mass balance rate of the selected 65 glaciers and their area

altitudes is high compared to higher altitudes. This is because the lower elevations are in high-temperature zone and higher altitudes are in lower temperature zone. It is also observed that there is a slight gain in mass at altitude band of 6200–6400 m, which may be because this altitude band is mostly covered by accumulation zones of some of the glaciers.

To quantify the improvements obtained due to DEM bias corrections applied in this study, the specific SMB rate for Chhota Shigri Glacier calculated with bias correction (-0.55 ± 0.26 m w.e.y^{-1}) and without bias corrections (0.18 ± 0.08 m w.e.y^{-1}) is compared with field values (-0.67 m w.e.y^{-1}) obtained from Azam et al. (2016). The comparison indicates that the error has reduced from 127 to 18%. Therefore, one can say that these corrections are very necessary for better estimation of glacier surface mass balance.

p.avinash283@gmail.com



Mass loss (Gt)*10^{-3}	6.7	40.2	120	216.5	344.5	573.6	923.9	1229.3	1165.3	326.9	64.3	3.8	-0.93
Area (km^2)	0.58	4.16	11.98	20.86	32.28	52.75	81.81	107.38	103.57	36.79	8.53	1.26	0.05

Fig. 5 Hypsometric distribution of ice loss in the selected glaciers of Chandra basin

Table 1 Comparison of the specific SMB rate obtained from the present study with other studies in Chandra basin

Study site	Specific SMB rate from the present study (1999–2013) (m w.e.y^{-1})	Specific SMB rate from other studies (m w.e.y^{-1})
Chhota Shigri Glacier	-0.55 ± 0.26	-0.46 ± 0.34[a]
Bara Shigri Glacier	-0.66 ± 0.32	-0.66 ± 0.31[a]
Samudra Tapu Glacier	-0.69 ± 0.33	-0.69 ± 0.31[a]
Hamtah Glacier	-0.55 ± 0.27	-0.45 ± 0.15[b]
Chandra Basin	-0.61 ± 0.29	-0.68 ± 0.15[c]; -0.65 ± 0.04[a]

[a]Vijay and Braun (2016) (2000–2012)
[b]Vincent et al. (2013) (1999–2011)
[c]Gardelle et al. (2013) (1999–2011)

Likewise, a comparison of the mass balance estimates between the present study and other studies given in Table 1. Table 1 indicates that the present study results closely match with the other studies. However, it is observed that the present study estimates for individual glaciers shown in Table 1 are slightly overestimated when compared with other studies. This is possibly due to the difference in the methodology used to correct the penetration of C-band. All other mentioned studies have used SRTM X-band data to correct the C-band penetration, whereas in this study the method given by Kääb et al. (2012) has been used.

p.avinash283@gmail.com

It is mainly because SRTM X-band data for the study area is not available but still other studies have used X-band data of nearby basin as a proxy for correction, which may not be appropriate. Specifically, when the results of Chhota Shigri Glacier (Table 1) are compared with the field-based glaciological estimates given by Azam et al. (2016), it is found that the present study results are closer than other studies. On the other hand, it is seen that the results of the whole Chandra basin are underestimated compared to other studies (Table 1). This is because the small glaciers of size <0.5 km^2 are not considered in our study.

5 Conclusion and Recommendations

The current study demonstrates that the geodetic mass balance modelling of glaciers after bias corrections is effective in estimating the long-term surface mass balance. Results of the current study show that the Chandra basin has lost 5.01 ± 2.48 Gt amount of ice mass during the course of 14 years. This is a clear signal indicating the effect of climate change occurred over the study area, which demonstrates how climate change causes imbalance in the health of glacier(s). However, such conclusions are based on very limited number of glaciers which have been studied under this research. Therefore, it is suggested that this modelling approach should be extended to other basins to get the current status of the health of the glaciers across Himalayas. This needs to be taken up on urgent basis considering the fact that many Himalayan glaciers are still untouched and needs to be investigated. In spite of bias corrections, the mass balance calculated from DEM differencing suffers from poor vertical accuracy of the satellite DEM data, thus restricting it to decadal scale studies only. Development of new satellite missions such as Resources at 3S, ICE Sat-2, CryoSat-2, Sentinel-1 can increase the DEM accuracy, providing better future prospects to understand health of the glaciers.

In order to study long-term effects of climate change on the health of glaciers located in the complex Himalayan terrain, continuous surface mass balance data on yearly and seasonal basis are also very much needed. Recent technologies such as Unmanned Aerial Vehicle (UAV), Terrestrial LiDAR are need to be explored which can provide a high-resolution and vertically accurate surface topographic information of the glaciers through which surface mass balance estimates can be obtained on seasonal as well as yearly basis. Such technology can be used to effectively monitor the benchmark glaciers which are presently surveyed using field techniques and also can be used to increase the number of benchmark glaciers studied.

Acknowledgements The lead author and the second author acknowledge the support provided by the Indian Institute of Technology Bombay, through Centre of Excellence in Climate Studies (IITB-CECS) project of the Department of Science and Technology (DST), New Delhi, India. All the authors are thankful to DLR, Germany for freely providing TanDEM-X CoSSC products under TanDEM-X Science proposal XTI_GLAC7043 as well as to USGS for freely providing the SRTM 30 m DEM.

p.avinash283@gmail.com

References

Arendt AA, Echelmeyer KA, Harrison WD, Lingle CS, Valentine VB (2002) Rapid wastage of Alaska glaciers and their contribution to rising sea level. Science 297(5580):382–386. https://doi.org/10.1126/science.1072497

Azam MF, Wagnon P, Ramanathan A, Vincent C, Sharma P, Arnaud Y, Linda A, Pottakkal JG, Chevallier P, Singh VB, Berthier E (2012) From balance to imbalance: a shift in the dynamic behaviour of Chhota Shigri glacier, western Himalaya, India. J Glaciol 58(208):315–324. https://doi.org/10.3189/2012JoG11J123

Azam MF, Ramanathan AL, Wagnon C, Vincent AL, Berthier E, Sharma P, Mandal A, Angchuk T, Singh VB, Pottakkal JG (2016) Meteorological conditions, seasonal and annual mass balances of Chhota Shigri Glacier, western Himalaya, India. Ann Glaciol 57(71):328–338. https://doi.org/10.3189/2016AoG71A570

Berthier E, Arnaud Y, Baratoux D, Vincent C, Rémy F (2004) Recent rapid thinning of the "Mer de Glace" glacier derived from satellite optical images. Geophys Res Lett 31(17). https://doi.org/10.1029/2004gl020706

Berthier E, Arnaud Y, Vincent C, Remy F (2006) Biases of SRTM in high-mountain areas: implications for the monitoring of glacier volume changes. Geophys Res Lett 33(8). https://doi.org/10.1029/2006gl025862

Berthier E, Arnaud Y, Kumar R, Ahmad S, Wagnon P, Chevallier P (2007) Remote sensing estimates of glacier mass balances in the Himachal Pradesh (Western Himalaya, India). Rem Sens Environ 108(3):327–338. https://doi.org/10.1016/j.rse.2006.11.017

Berthier E, Schiefer E, Clarke GK, Menounos B, Rémy F (2010) Contribution of Alaskan glaciers to sea-level rise derived from satellite imagery. Nat Geosci 3(2):92–95. https://doi.org/10.1038/ngeo737

Bhutiyani MR, Kale VS, Pawar NJ (2007) Long-term trends in maximum, minimum and mean annual air temperatures across the Northwestern Himalaya during the twentieth century. Clim Change 85(1–2):159–177. https://doi.org/10.1007/s10584-006-9196-1

Bolch T, Kulkarni A, Kääb A, Huggel C, Paul F, Cogley JG, Frey H, Kargel JS, Fujita K, Scheel M, Bajracharya S (2012) The state and fate of Himalayan glaciers. Science 336(6079):310–314. https://doi.org/10.1126/science.1215828

Bookhagen B, Burbank DW (2010) Toward a complete Himalayan hydrological budget: spatiotemporal distribution of snowmelt and rainfall and their impact on river discharge. J Geophys Res Earth Surf 115(F3). https://doi.org/10.1029/2009jf001426

Dash SK, Jenamani RK, Kalsi SR, Panda SK (2007) Some evidence of climate change in twentieth-century India. Clim Change 85(3):299–321. https://doi.org/10.1007/s10584-007-9305-9

Dedieu JP, Rabatel A, Vincent C, Valla F, Thibert E, Arnaud Y (2003) Glacier mass balance determination by remote sensing in the French Alps: progress and limitation for time series monitoring. In: Proceedings of Geoscience and remote sensing symposium, 2003. IGARSS'03, vol 4, pp 2602–2604. https://doi.org/10.1109/igarss.2003.1294523

Dimri AP, Dash SK (2010) Winter temperature and precipitation trends in the Siachen Glacier. Curr Sci, 1620–1625

Dimri AP, Dash SK (2012) Wintertime climatic trends in the western Himalayas. Clim Change 111(3–4):775–800. https://doi.org/10.1007/s10584-011-0201-y

Dobhal DP, Gergan JT, Thayyen RJ (2004) Recession and morphogeometrical changes of Dokriani glacier (1962-1995) Garhwal Himalaya, India. Curr Sci 86(5):692–696

DST (2012) Dynamics of glaciers in the Indian Himalaya: science plan. Prepared by R.K. Midha. Published by the Science and Engineering Board, Department of Science and Technology, New Delhi, Himalayan Glaciology Technical Report No. 2, p 125. Accessed 3 Oct 2017

Gardelle J, Berthier E, Arnaud Y (2012) Slight mass gain of Karakoram glaciers in the early 21st century. Nat Geosci 5:322–325. https://doi.org/10.1038/ngeo1450

Gardelle J, Berthier E, Arnaud Y, Kääb A (2013) Region-wide glacier mass balances over the Pamir-Karakoram-Himalaya during 1999–2011. The Cryosphere 7(4):1263–1286

p.avinash283@gmail.com

Gautam R, Christina Hsu N, Lau K.-M (2010) Premonsoon aerosol characterization and radiative effects over the Indo-Gangetic Plains: Implications for regional climate warming. Journal of Geophysical Research 115(D17)

Houghton JT, Ding Y, Griggs DJ, Noguer M, van der Linden PJ, Dai X, Maskell K, Johnson CA (2001) Technical summary of working group 1. In: Climate change 2001: the scientific basis. Contributions of working group I to the third assessment report of the intergovernmental panel on climate change. Cambridge University Press. Accessed 15 Nov 2017

Huss M (2013) Density assumptions for converting geodetic glacier volume change to mass change. Cryosphere Discuss 7(1):219–244. https://doi.org/10.5194/tcd-7-219-2013

Immerzeel WW, Van Beek LP, Bierkens MF (2010) Climate change will affect the Asian water towers. Science 328(5984):1382–1385. https://doi.org/10.1126/science.1183188

IPCC (2007) Summary for policymakers. In: Solomon S, Qin D, Manning M, Chen Z, Marquis M, Averyt KB, Tignor M, Miller HL (eds) Climate change 2007: the physical science basis. Contribution of working group I to the fourth assessment report of the intergovernmental panel on climate change. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA. Available at https://www.ipcc.ch/pdf/assessment-report/ar4/wg1/ar4-wg1-spm.pdf. Accessed 02 Nov 2017

IPCC (2014) Climate change 2014: synthesis report. In: Pachauri RK, Meyer LA (eds) Contribution of working groups I, II and III to the fifth assessment report of the intergovernmental panel on climate change. IPCC, Geneva, Switzerland, p 151. Available at https://www.ipcc.ch/pdf/assessmentreport/ar5/syr/AR5_SYR_FINAL_SPM.pdf. Accessed 07 Dec 2017

Kääb A (2007) Glacier volume changes using ASTER optical stereo. A test study in Eastern Svalbard. In: Geoscience and remote sensing symposium, 2007. IGARSS 2007, pp 3994–3996. https://doi.org/10.1109/igarss.2007.4423724

Kääb A, Berthier E, Nuth C, Gardelle J, Arnaud Y (2012) Contrasting patterns of early twenty-first-century glacier mass change in the Himalayas. Nature 488(7412):495–498. https://doi.org/10.1038/nature11324

Krimmel RM (1999) Analysis of difference between direct and geodetic mass balance measurements at South Cascade Glacier, Washington. Geografiska Annaler Ser A Phys Geogr 81(4):653–658. https://doi.org/10.1111/1468-0459.00093

Kulkarni AV, Bahuguna IM, Rathore BP, Singh SK, Randhawa SS, Sood RK, Dhar S (2007) Glacial retreat in Himalaya using Indian Remote Sensing satellite data. Curr Sci, 69–74

Kumar R, Areendran G, Rao P (2009) Witnessing change: glaciers in the Indian Himalayas. Pilani, WWF-India and Birla Institute of Technology. Available at http://awsassets.wwfindia.org/downloads/glacier_report.pdf. Accessed 10 Dec 2017

Mernild SH, Pelto M, Malmros JK, Yde JC, Knudsen NT, Hanna E (2013) Identification of snow ablation rate, ELA, AAR and net mass balance using transient snowline variations on two Arctic glaciers. J Glaciol 59(216):649–659. https://doi.org/10.3189/2013JoG12J221P

Muskett RR, Lingle CS, Tangborn WV, Rabus BT (2003) Multi-decadal elevation changes on Bagley Ice Valley and Malaspina Glacier, Alaska. Geophys Res Lett 30(16)

Muskett RR, Lingle CS, Sauber JM, Post AS, Tangborn WV, Rabus BT, Echelmeyer KA (2009) Airborne and spaceborne DEM-and laser altimetry-derived surface elevation and volume changes of the bering glacier system, Alaska, USA, and Yukon, Canada, 1972–2006. J Glaciol 55(190):316–326. https://doi.org/10.3189/002214309788608750P

Narama C, Kääb A, Kajiura T, Abdrakhmatov K (2009) Spatial variability of recent glacier area changes in the Tien Shan Mountains, Central Asia, using Corona (~1970), Landsat (~2000), and ALOS (~2007) satellite data. Glob Plan Change 71:42–54. https://doi.org/10.1016/j.gloplacha.2009.08.002

Negi HS, Saravana G, Rout R, Snehmani (2013) Monitoring of great Himalayan glaciers in Patsio region, India using remote sensing and climatic observations. Curr Sci, 1383–1392

Nuth C, Kääb A (2011) Co-registration and bias corrections of satellite elevation data sets for quantifying glacier thickness change. Cryosphere 5(1):271. https://doi.org/10.5194/tc-5-271-2011

p.avinash283@gmail.com

Østrem G, Haakensen N (1999) Map comparison or traditional mass-balance measurements: which method is better? Geografiska Annaler Ser A Phys Geogr 81(4):703–711. https://doi.org/10.1111/1468-0459.00098

Pandey P, Ali SN, Ramanathan AL, Venkataraman G (2017) Regional representation of glaciers in Chandra Basin region, western Himalaya, India. Geosci Front. https://doi.org/10.1016/j.gsf.2016.06.006

Pandit A, Ramsankaran R, Rao YS (2014) Generation and validation of the interferometric SAR DEMs from TanDEM-X data for Gangotri and Hamtah Glaciers of Indian Himalayas. Procedia Technol 16:793–805. https://doi.org/10.1016/j.protcy.2014.10.029

Paul F (2008) Calculation of glacier elevation changes with SRTM: is there an elevation-dependent bias? J Glaciol 54(188):945–946. https://doi.org/10.3189/002214308787779960

Rabatel A, Dedieu JP, Vincent C (2016) Spatio-temporal changes in glacier-wide mass balance quantified by optical remote sensing on 30 glaciers in the French Alps for the period 1983–2014. J Glaciol 62(236):1153–1166. https://doi.org/10.1017/jog.2016.113

Racoviteanu AE, Manley WF, Arnaud Y, Williams MW (2007) Evaluating digital elevation models for glaciologic applications: an example from Nevado Coropuna, Peruvian Andes. Glob Plan Change 59(1):110–125. https://doi.org/10.1016/j.gloplacha.2006.11.036

Ramsankaran R, Pandit A, Azam MF (2018) Spatially distributed ice-thickness modelling for Chhota Shigri Glacier in western Himalayas, India. Int J Rem Sens (in press)

Rignot E, Rivera A, Casassa G (2003) Contribution of the Patagonia Icefields of South America to sea level rise. Science 302(5644):434–437. https://doi.org/10.1126/science.1087393

Rivera A, Casassa G (1999) Volume changes on Pio XI glacier, Patagonia: 1975–1995. Glob Plan Change 22(1):233–244. https://doi.org/10.1016/S0921-8181(99)00040-5

Rolstad C, Haug T, Denby B (2009) Spatially integrated geodetic glacier mass balance and its uncertainty based on geostatistical analysis: application to the western Svartisen ice cap, Norway. J Glaciol 55(192):666–680. https://doi.org/10.3189/002214309789470950

Sangewar CV, Shukla SP, Singh RK (2009) Inventory of the Himalayan Glaciers—a contribution to international hydrological Programme. Special Publication No. 34, Geological Survey of India

Schiefer E, Menounos B, Wheate R (2007) Recent volume loss of British Columbian glaciers, Canada. Geophys Res Lett 34(16). https://doi.org/10.1029/2007gl030780

Sharma SS, Ganju A (2000) Complexities of avalanche forecasting in Western Himalaya—an overview. Cold Reg Sci Technol 31(2):95–102. https://doi.org/10.1016/S0165-232X(99)00034-8

Sharma P, Ramanathan AL, Pottakkal J (2013) Study of solute sources and evolution of hydrogeochemical processes of the Chhota Shigri Glacier meltwaters, Himachal Himalaya, India. Hydrol Sci J 58(5):1128–1143. https://doi.org/10.1080/02626667.2013.802092

Sharma P, Patel LK, Ravindra R, Singh A, Mahalinganathan K, Thamban M (2016) Role of debris cover to control specific ablation of adjoining Batal and Sutri Dhaka glaciers in Chandra Basin (Himachal Pradesh) during peak ablation season. J Earth Syst Sci 125(3):459–473. https://doi.org/10.1007/s12040-016-0681-2

Singh SP, Bassignana-Khadka I, Karky BS, Sharma E (2011) Climate change in the Hindu Kush-Himalayas: the state of current knowledge. International Centre for Integrated Mountain Development (ICIMOD). Available at: http://lib.icimod.org/record/9417/files/icimod-climate_change_in_the_hindu_kush-himalayas.pdf. Accessed 5 Dec 2017

Sund M, Eiken T, Hagen JO, Kääb A (2009) Svalbard surge dynamics derived from geometric changes. Ann Glaciol 50(52):50–60. https://doi.org/10.3189/172756409789624265

Surazakov AB, Aizen VB (2006) Estimating volume change of mountain glaciers using SRTM and map-based topographic data. IEEE Trans Geosci Remote Sens 44(10):2991–2995. https://doi.org/10.1109/TGRS.2006.875357

Van Niel TG, McVicar TR, Li LT, Gallant JC, Yang QK (2008) The impact of 1394 misregistration on SRTM and DEM image differences. Rem Sens Environ, 1395 112(5):2430–2442. https://doi.org/10.1016/j.rse.2007.11.003

p.avinash283@gmail.com

Vijay S, Braun M (2016) Elevation change rates of glaciers in the Lahaul-Spiti (Western Himalaya, India) during 2000–2012 and 2012–2013. Rem Sens 8(12):1038. https://doi.org/10.3390/rs8121038

Vincent C, Ramanathan A, Wagnon P, Dobhal DP, Linda A, Berthier E, Sharma P, Arnaud Y, Azam MF, Gardelle J (2013) Balanced conditions or slight mass gain of glaciers in the Lahaul and Spiti region (northern India, Himalaya) during the nineties preceded recent mass loss. Cryosphere 7(2):569–582. https://doi.org/10.5194/tc-7-569-2013

Wagnon P, Linda A, Arnaud Y, Kumar R, Sharma P, Vincent C, Pottakkal JG, Berthier E, Ramanathan A, Hasnain SI, Chevallier P (2007) Four years of mass balance on Chhota Shigri Glacier, Himachal Pradesh, India, a new benchmark glacier in the western Himalaya. J Glaciol 53(183):603–611. https://doi.org/10.3189/002214307784409306

p.avinash283@gmail.com